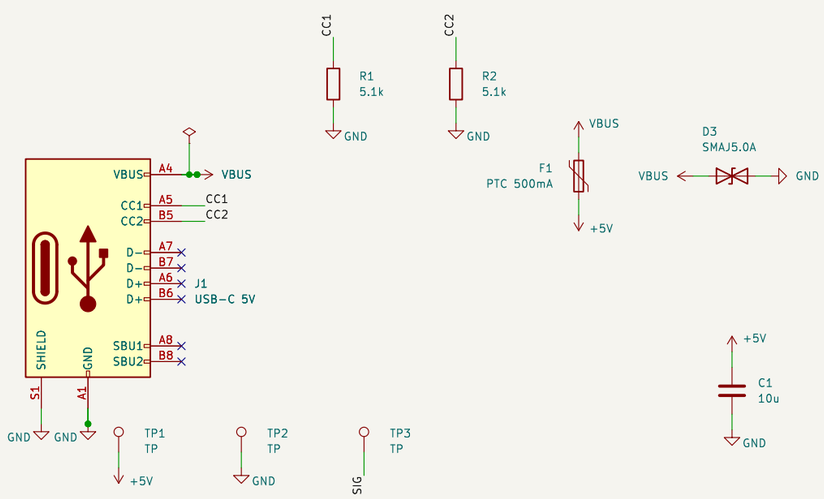
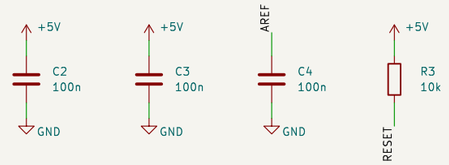
USB–C 5 V INPUT: CC pull–downs, PTC, TVS

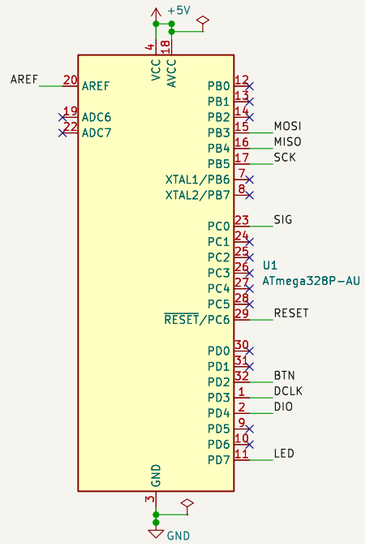


USER: reset button, LED

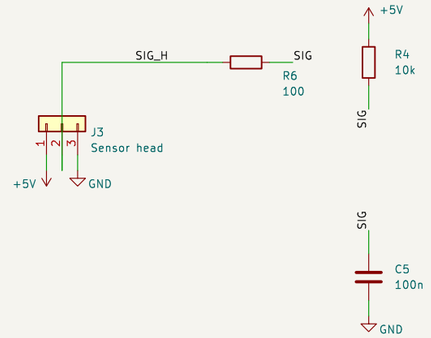


MCU: ATmega328P–AU, internal RC 8 MHz

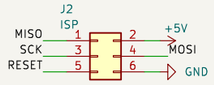




METER SENSOR HEAD INTERFACE



PROGRAMMING / DISPLAY





USB–C power only: 5.1k on CC1/CC2 advertises a 5 V sink; D+/D– unused. PTC limits a fault to 0.5 A, SMAJ5.0A clamps spikes.

Head: phototransistor between SIG_H and GND, 10k pull–up R4: SIG falls when the meter LED lights. R6 100R + C5 100n filter the cable.

ATmega328P at 5 V may run at 8 MHz (internal RC, fuses lfuse 0xE2 hfuse 0xD9 efuse 0xFD).

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /
File: bijli_meter_reader.kicad_sch

Title: Bijli Meter Reader — main board

Size: A3	Date: 2026–11–01	**Rev: B**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/1